

JKBOSE CLASS 10TH - MATHEMATICS

Series: 10th ARM(HZ) JKUTL 2024-25 - Series C-3-X | Code: 403-X | Total Questions: 40

Time: 3 Hours

Max Marks: 80 Marks

GENERAL INSTRUCTIONS: Hard Zone Annual Regular Examination. Section A (1m), B (2m), C (3m), D (4m).**SECTION - A : OBJECTIVE & VERY SHORT QUESTIONS**

(1 Mark Each)

1. In an A.P., if $d = -4$, $n = 7$, and $a_n = 4$, then the first term 'a' is: [1 Mark]
(A) 6
(B) 7
(C) 20
(D) None of these
2. The distance of the point $P(2, 3)$ from the X-axis is: [1 Mark]
(A) 2
(B) 3
(C) 1
(D) None of these
3. Prime factors of 156 are: [1 Mark]
(A) $2^2 \times 3 \times 13$
(B) $2^3 \times 3 \times 13$
(C) $2^1 \times 3 \times 13$
(D) None of these
4. For what value of K do the equations $3x - y + 8 = 0$ and $6x - Ky + 16 = 0$ represent coincident lines? [1 Mark]
(A) $1/2$
(B) $-1/2$
(C) 2
(D) None of these
5. If triangle ABC is right angled at C, then the value of $\cos(A + B)$ is: [1 Mark]
(A) 0
(B) 1
(C) $1/2$
(D) None of these
6. Factors of the quadratic equation $y^2 - 5y + 6 = 0$ are: [1 Mark]
(A) $(y - 2)(y - 3) = 0$
(B) $(y - 2)(y - 4) = 0$
(C) $(y - 5)(y - 1) = 0$
(D) None of these
7. In the standard form of a quadratic polynomial $ax^2 + bx + c$, the coefficients a, b, c are: [1 Mark]
(A) All are real numbers
(B) All are rational numbers
(C) a is non-zero real number, b and c are any real numbers
(D) None of these

8. A card is drawn from a pack of 52 cards. The probability of drawing a red face card is: [1 Mark]
(A) $2 / 13$
(B) $3 / 26$
(C) $1 / 13$
(D) None of these
9. Fundamental theorem of Arithmetic states that every composite number can be expressed as a of prime factors. [1 Mark]
(A) Sum
(B) Product
(C) Difference
(D) None of these
10. If we change the shape of an object from a sphere to a cylinder, then the volume of the cylinder will: [1 Mark]
(A) Increase
(B) Decrease
(C) Remain unchanged
(D) None of these
11. If $P(\text{not } E) = 0.5$, then the value of $P(E) = 1$. (State True or False) [1 Mark]
12. The sum of the first 22 terms of the A.P. 8, 3, -2, ... is: [1 Mark]
(A) -978
(B) -977
(C) -900
(D) None of these
13. All circles are (Congruent / Similar) [1 Mark]
14. The common point of the tangent and the circle is called the [1 Mark]
15. Write the general form of an Arithmetic Progression (A.P.). [1 Mark]
16. The linear equations $2x + 3y = 9$ and $4x + 6y = 18$ have infinitely many solutions. (State True or False) [1 Mark]
17. The value of $\sin \theta$ increases as θ increases from 0 deg to 90 deg. (State True or False) [1 Mark]
- OR ---
- Evaluate: $\sin^2 50^\circ + \cos^2 50^\circ = \dots\dots\dots$
18. The L.C.M. and H.C.F. of 6 and 20 are respectively: [1 Mark]
(A) 2, 60
(B) 60, 2
(C) 2, 30
(D) None of these
19. Calculate the mean of the first 6 prime numbers. [1 Mark]
20. Write the empirical relationship between mean, median and mode. [1 Mark]

Define the mode of grouped data.

--- OR ---

SECTION - B : SHORT ANSWER QUESTIONS

(2 Marks Each)

21. State whether the pair of linear equations $2x - 3y = 8$ and $4x - 6y = 9$ are consistent or inconsistent. [2 Marks]
22. Find the discriminant of the quadratic equation $9x^2 - 6x + 1 = 0$ and hence find the nature of its roots. [2 Marks]
23. In a right triangle ABC, right-angled at B, if $\tan A = 1$, then verify that $2 \sin A \cos A = 1$. [2 Marks]
24. Find the total surface area of a hemisphere whose radius is 21 cm. [2 Marks]

--- OR ---

Calculate the curved surface area of a cone whose slant height is 21 cm and diameter of its base is 24 cm.

25. Show that (4, 5), (7, 6), (4, 3) and (1, 2) are the vertices of a parallelogram. [2 Marks]

--- OR ---

Find a point on the Y-axis which is equidistant from the points A(6, 5) and B(-4, 3).

26. Find the zeroes of the quadratic polynomial $2x^2 - 8x + 6$. [2 Marks]

SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

27. If A and B are (-2, -2) and (2, -4) respectively, find the coordinates of P such that $AP = \frac{3}{7} AB$ and P lies on AB. [3 Marks]
28. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes. [3 Marks]
29. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. [3 Marks]

--- OR ---

XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that angle AOB = 90 deg.

30. State and prove the converse of Basic Proportionality Theorem (Thales' Theorem). [3 Marks]
31. In the given figure, $QR / QS = QT / PR$ and angle 1 = angle 2. Prove that Triangle PQS ~ Triangle TQR. [3 Marks]
32. Given that $\sqrt{7}$ is an irrational number, prove that $5 + 2\sqrt{7}$ is an irrational number. [3 Marks]

33. In the following situation, does the list of numbers make an A.P. and why? 'The taxi fare after each km when the fare is Rs 15 for the first km and Rs 8 for each additional km.'

[3 Marks]

--- OR ---

If the sum of the first n terms of an A.P. is $4n - n^2$, what is the first term (S_1)? What is the sum of the first two terms? What is the second term? Similarly find the 3rd, 10th and n th terms.

34. Two dice (one blue and one grey) are thrown at the same time. Find the probability that the sum of the numbers appearing on top is: (i) 8, (ii) less than or equal to 12.

[3 Marks]

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

[4 Marks]

--- OR ---

Is it possible to design a rectangular mango grove whose length is twice its breadth and the area is 800 m^2 ? If so, find its length and breadth.

36. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and total height of the vessel is 13 cm. Find the inner surface area of the vessel.

[4 Marks]

--- OR ---

A spherical glass vessel has a cylindrical neck 8 cm long, 2 cm in diameter; the diameter of the spherical part is 8.5 cm. By measuring the amount of water it holds, a child finds the volume to be 345 cm^3 . Check whether she is correct (taking $\pi = 3.14$).

37. Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles.

[4 Marks]

38. Write all other trigonometric ratios of angle A in terms of $\sec A$.

[4 Marks]

--- OR ---

Prove the trigonometric identity: $[\cos A / (1 + \sin A)] + [(1 + \sin A) / \cos A] = 2 \sec A$.

39. If in two triangles, corresponding angles are equal, then prove that their corresponding sides are in the same ratio and hence the two triangles are similar.

[4 Marks]

--- OR ---

The diagonals of a quadrilateral ABCD intersect each other at O such that $AO / BO = CO / DO$. Show that ABCD is a trapezium.

40. The distribution given below gives the weights of 30 students of a class. Find the median weight of the students:

[4 Marks]

Weight (in kg)	Number of Students
40 - 45	2
45 - 50	3
50 - 55	8
55 - 60	6
60 - 65	6
65 - 70	3
70 - 75	2
Total	30